

Day 2 — Machine- Learning Approaches

Objectives

- Build unsupervised detectors that learn boundaries of normal behavior.
- Engineer compact window/lag features with leakage-safe validation.
- Train Isolation Forest (core) and One-Class SVM / LOF.

Feature Engineering (Univariate/Per-Metric)

- Window stats: mean, std, min/max, median, MAD, rolling slopes.
- Lag features: $t - 1$, $t - 2$, seasonal lags ($t - 24h$, $t - 7d$ where applicable).
- Frequency domain: energy in bands, spectral entropy (optional).
- Leakage guards: train/val split by time; avoid shuffling; fit scalers on train only.

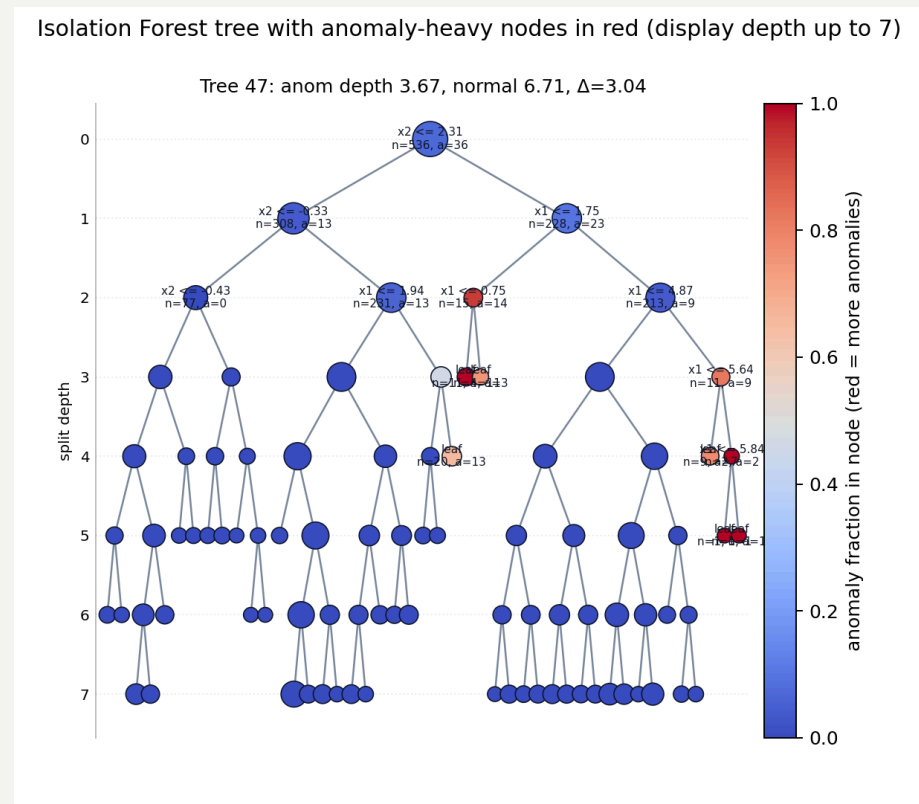
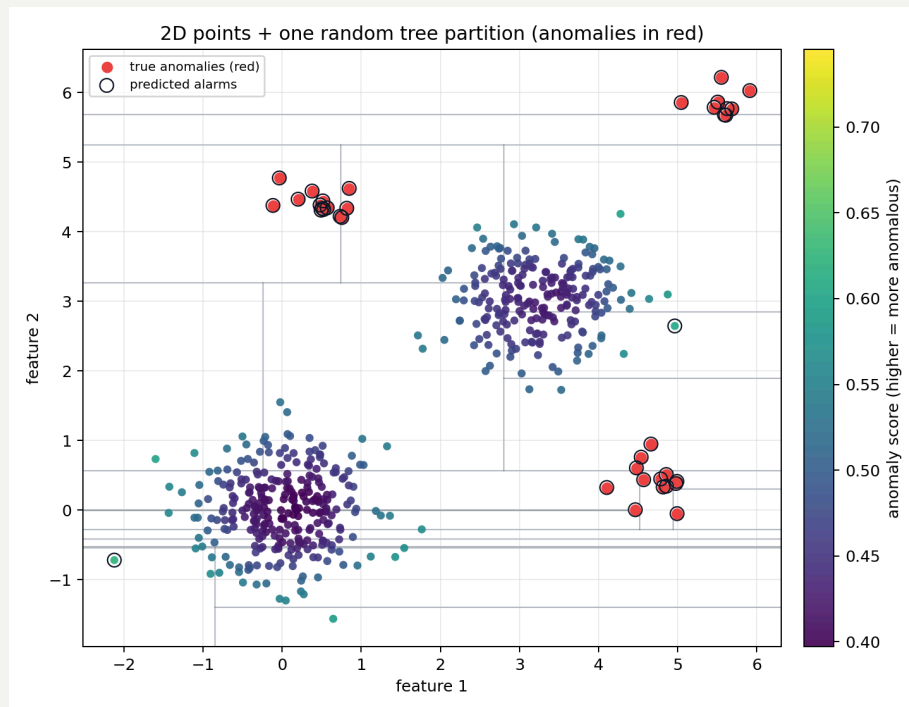
Isolation Forest

- Idea: anomalies tend to isolate in fewer splits on average across many random trees.
- Why: outliers lie in sparse regions, so random cuts separate them earlier than dense normal clusters.
- **Not Random Forest training:** splits are not impurity-optimized; feature + threshold are chosen randomly.

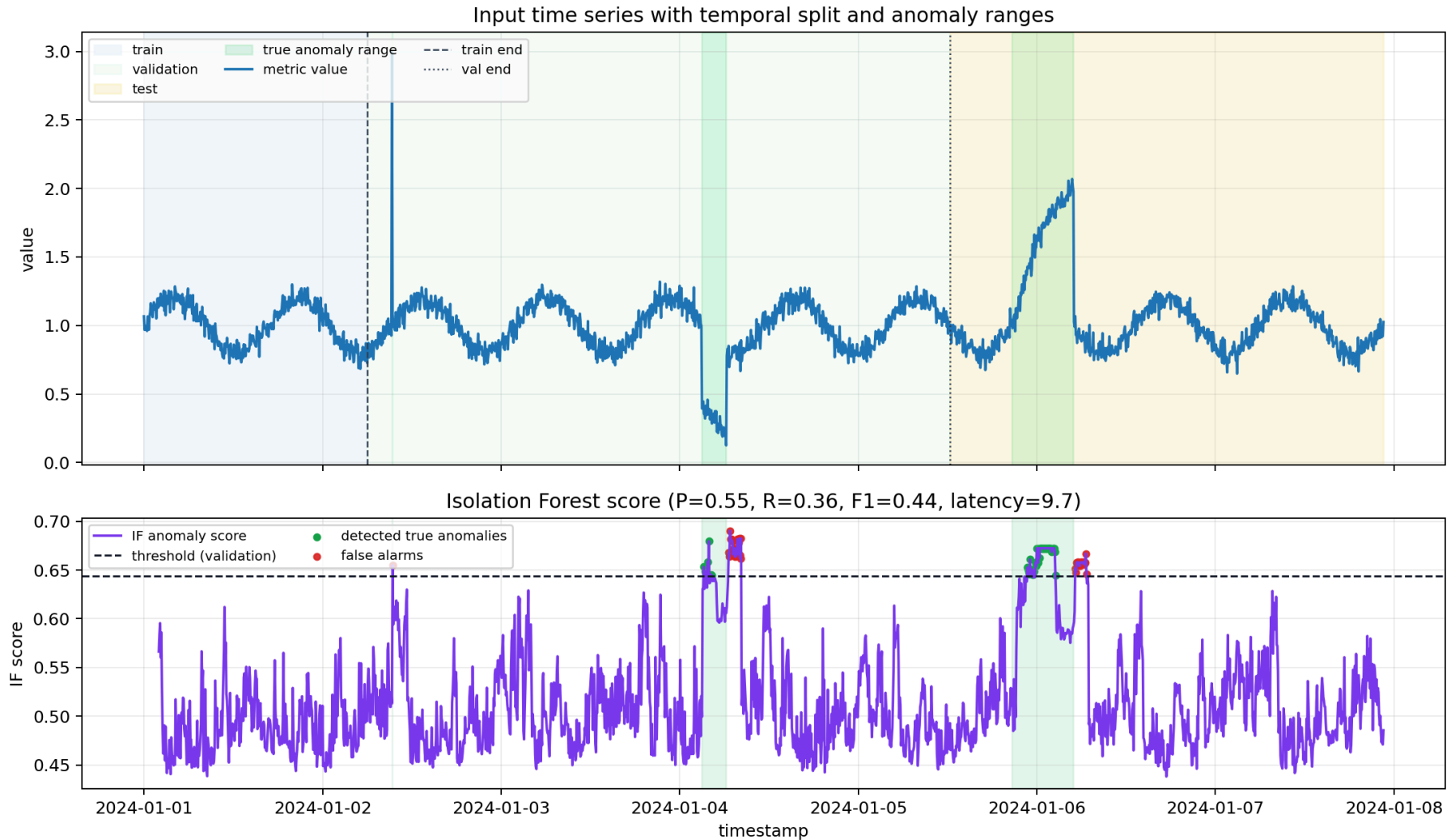
Key parameters:

Parameter	Explanation
<code>n_estimators</code>	Number of trees in the ensemble.
<code>max_samples</code>	Number (or fraction) of samples per tree. Smaller subsamples increase randomness; common default is <code>min(256, n_samples)</code> .
<code>contamination</code>	Bias on the final labeling rate.
<code>max_features</code>	Number of features considered when splitting. Lower values increase tree diversity in high-dimensional data.

- Scores: based on average path length across trees; invert sign so higher = more anomalous.
- Advantages: non-parametric, handles non-linearities, linear complexity, low memory.
- time-aware by itself; for time series you must use lag/window features and time-based train/val/test splits.
- Common failure modes: collective anomalies that look locally dense, and drift (needs periodic refit/recalibration).



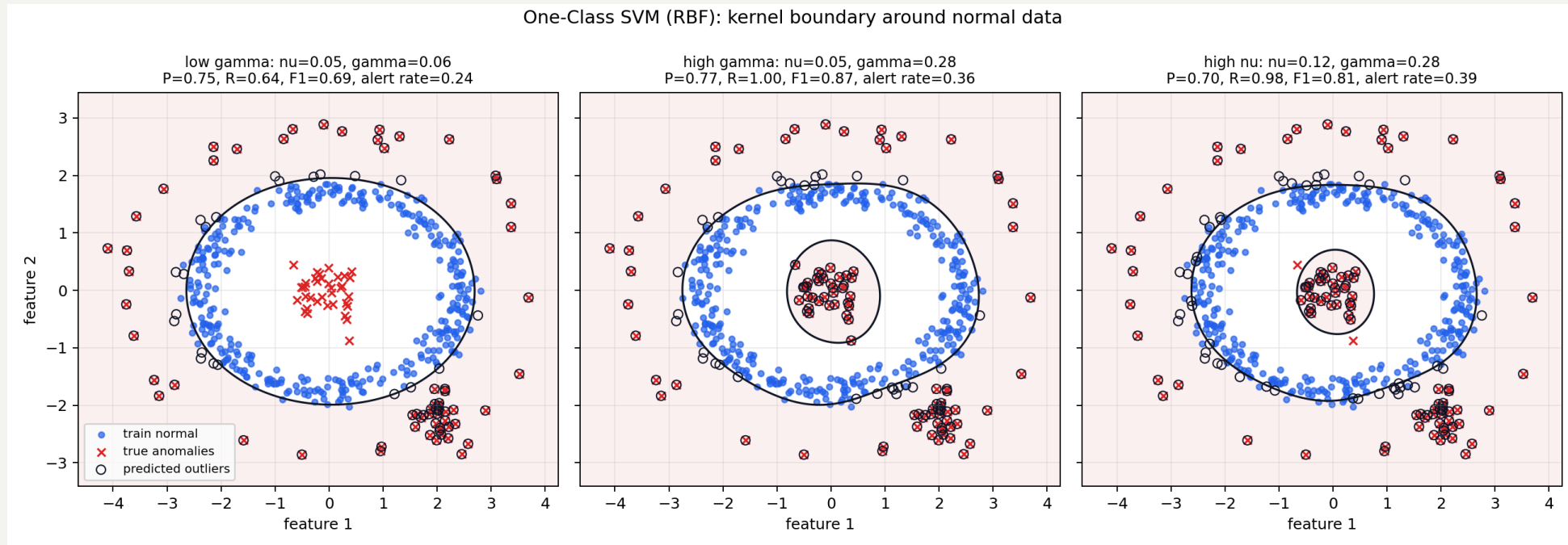
IF Score Over Time



OCSVM / LOF

- One-Class SVM: kernel boundary around normal; choose `nu` (upper bound on anomalies) and `gamma` for RBF.
- Local Outlier Factor: density-based; good for local deviations; needs neighborhood size `n_neighbors`.
- Use when: density shifts or local patterns dominate.

OCSVM Kernel Boundary (Visual)



LOF Local Density (Visual)

